

Wind pressures on roofs of nonrectangular buildings: Experimental and machine learning approaches

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Abstract. Wind effects on buildings with rectangular plans have been investigated widely by wind engineering researchers either in wind tunnels or through CFD simulations. These studies provided comprehensive overviews and detailed descriptions of wind pressures on rectangular buildings and created the basic source required to formulate the wind provisions in the national and international codes and standards. However, buildings with irregular (i.e., non-rectangular) plans have not received adequate attention from wind tunnel investigations. Therefore, wind loads on irregular buildings are described shortly and shyly, if at all, in the current building codes and standards. This paper describes the experimental investigations into the flat-roof pressures of buildings with four non-rectangular shapes —L, U, T, and X— in an atmospheric boundary layer wind tunnel. The results reveal that the distribution of wind loads on the outer roof corners and edges of buildings with non-rectangular plans resembles that experienced by a rectangular building. However, the wind loads on the inner perimeter area, particularly the inner edge of the U-shaped building, were observed to be generally higher than those recorded on the edges of a typical rectangular roof. Furthermore, the wind tunnel measurements not only provided valuable data but also served as a dataset when applying Machine Learning (ML) as a tool to predict wind loads on irregular buildings. This involved the utilization of a Gradient Boosting Regressor (GBR) and Artificial Neural Networks (ANN), using two data split approaches: random and structured splits. The ML models exhibit significant predictive accuracy, achieving minimal Mean Squared Error (MSE) and coefficients of determination (R-squared) of about 0.97 for wind pressure coefficients. Further, the study demonstrated that a structured split of the dataset reflects a more realistic assessment of the ML models.

Keywords: machine learning; nonrectangular buildings; pressure zonal system; random split; roof pressures; structured split; wind tunnel testing

1. Introduction

Most buildings worldwide have a plan of a rectangle; however, buildings with an irregular plan, non-rectangular buildings, exist. A survey was conducted by the authors on four cities in Canada, Montreal, Longueuil, Ottawa, and Vancouver, to know the number of buildings with non-rectangular plans and the percentage of such buildings in the urban areas. Using Google Earth, specific areas were considered in each of the four cities, and the buildings were counted manually block by block in the chosen areas.

The survey indicated that there exists a significant number of irregular buildings in urban cities, 1400 buildings in four cities, and irregular buildings form 8% of the existing buildings as shown in Fig. 1. Fig. 2 categorizes the irregular buildings based on shape, e.g., L-shaped and T-shaped buildings; it was found that L-shaped buildings are the most common. The complex shape category refers to buildings with non-common shapes in Fig. 2.

Wind design standards comprehensively address the wind design of buildings with rectangular plans. Conversely, the wind design of buildings with irregular

(non-rectangular) plans is addressed through inadequate and brief guidelines in wind provisions. Numerous wind studies have investigated wind loads on roofs and walls of non-rectangular buildings using wind tunnels (WT) and Computational Fluid Dynamics (CFD), as evidenced by several research studies (e.g., Cook 1990, Jamieson *et al.* 1992, Stathopoulos and Zhou 1993 Morse 2003, Gomes *et al.* 2005, Parackal *et al.* 2016, Shao *et al.* 2018, Shao *et al.* 2019, Matus and Zisis 2021, Sarma *et al.* 2023, Verma and Sonparote 2023, Aldoum and Stathopoulos 2024a and b, Van Den Heuvel and Kopp 2024, and Aldoum and Stathopoulos 2025)

Jamieson *et al.* (1992) reported wind peak pressure coefficients on walls experimentally at 2/3 of the building height. This does not reflect the most critical suction loads, which should be considered in the building design. Stathopoulos and Zhou (1993) evaluated mean wind loads numerically. Cook (1990) suggested a method to evaluate wind loads on nonrectangular buildings based on data of rectangular buildings.

Gomes *et al.* (2005) tested irregular building models in a uniform wind tunnel that doesn't simulate the atmospheric boundary layer. Shao *et al.* (2019) provided insightful details on wind loads on hipped roof buildings, the study included mean, peak, and area-averaged wind loads. Aldoum and Stathopoulos (2025) compared roof pressure coefficients of nonrectangular buildings, obtained through

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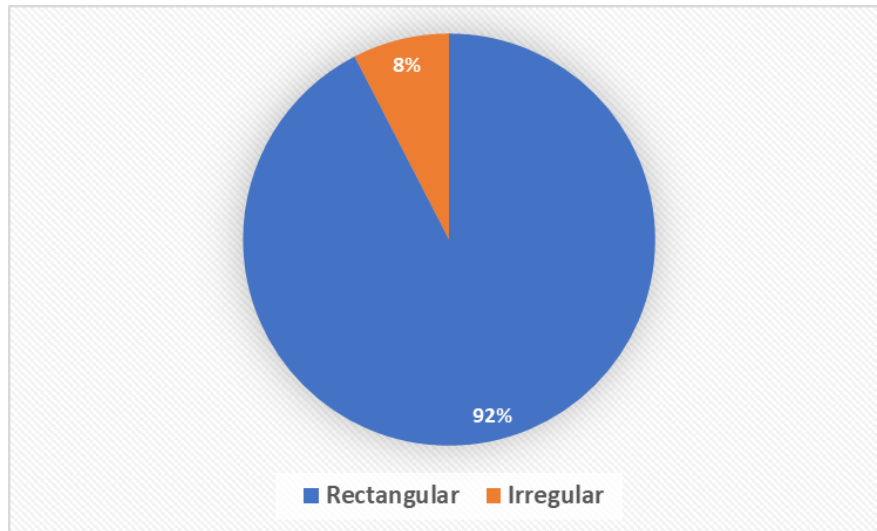


Fig. 1 Percentage of existing irregular rectangular buildings in the four cities based on the survey

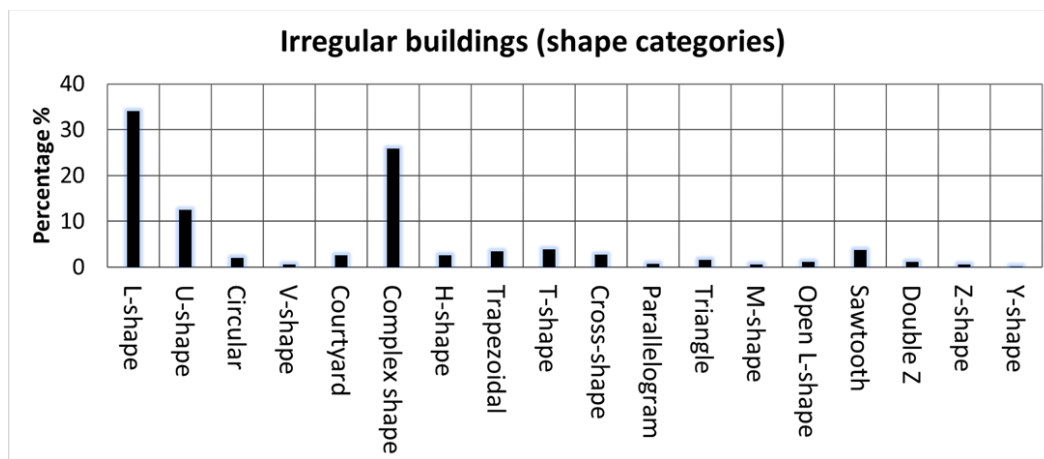


Fig. 2 Categories of irregular buildings based on shape

wind tunnel testing, with the provisions of ASCE 7-22 and NBCC 2020.

Utilizing Machine Learning (ML) offers a potent means to conduct multiple and multivariate regression analyses on intricate datasets. In wind engineering, ML has found application in predicting wind loads, wind speeds, and various other topics within the field. Khanduri *et al.* (1997) provides a generalized model for the evaluation of the wind loads on non-isolated buildings using a neural network (NN), the study used a small dataset for the training of the NN to predict the ratio of base overturning moment of buildings with surroundings and isolated buildings, interference factor (IF), trained NN exhibited a good generalization for along wind mean IF values when was compared testing set. Bitsuamlak *et al.* (2007) employed a neural network to forecast speed-up ratios for diverse topographic features such as escarpments and hills. Elshaer *et al.* (2017) Alanani and Elshaer (2023) utilized Artificial Neural Networks (ANN) in optimizing the aerodynamics corners in tall buildings and the layout of shear walls.

ML was implemented to model wind loads on buildings with rectangular plans (Gavalda *et al.* 2011, Dongmei *et al.* 2017, Bre *et al.* 2018, Fernández-Cabán *et al.* 2018, Tian *et al.*

al. 2020, Hu *et al.* 2020, Kim *et al.* 2021, Sang *et al.* 2021, Lamberti and Gorié 2021, Ding *et al.* 2022, Weng and German Paal 2022, Meddage *et al.* 2022, Chen *et al.* 2022, Du and Chen 2023, Meddage *et al.* 2024, and Tong *et al.* 2024).

Bre *et al.* (2018) constructed three ANNs to predict surface average pressure coefficients (C_p) on walls and roofs of flat, gabled, and hipped roof low-rise buildings, using the TPU database. Higgins and Stathopoulos (2021) built ANNs to predict wind speeds in urban areas to find the optimal location of wind turbines. The ANN showed excellent results with high success rates. Fernández-Cabán *et al.* (2018) employed a feed-forward ANN with backpropagation to predict mean, root-mean-square (rms), and peak pressure coefficients on roofs of low-rise building models with different geometric scales and approaching flow conditions characterized by a range of turbulence intensities. The study used the turbulence intensity and the roof taps coordinates as inputs for the ANN.

Tian *et al.* (2020) used ANNs to generalize wind pressure coefficients on low-rise gabled roof buildings, leveraging NIST experimental data. The study used the x, y, and z coordinates and the wind direction as input features.

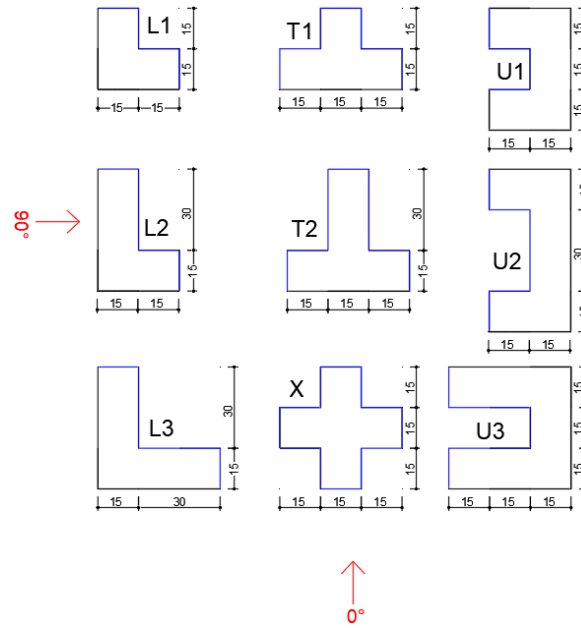


Fig. 3 Plans of the tested models with the reference wind directions, dimensions in meters (full-scale)

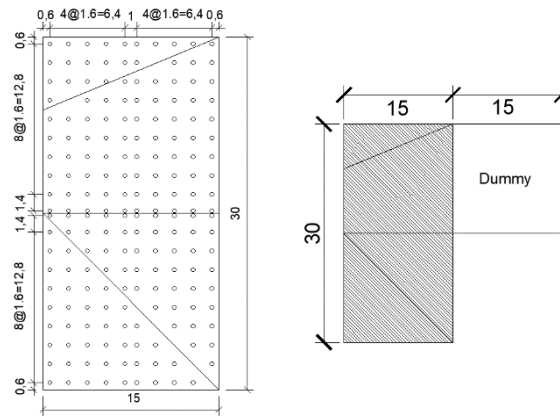


Fig. 4 Pressure tap layout, and the basic and dummy models combined to form an L-shaped model (dimensions in meters, full-scale)

Sang *et al.* (2021) evaluated wind drag coefficients on high-rise rectangular buildings numerically, the study used the numerical results to train and test ANNs. Lamberti and Gorlé (2021) employed Large-Eddy Simulations (LES) and Reynolds-Averaged Navier-Stokes (RANS) to assess wind pressures on a 100-meter-tall building. The data generated from these simulations was then leveraged to predict root-mean-square (rms) pressure coefficients using an ANN. Du and Chen (2023) used a semi-supervised regression model to predict peak wind pressure coefficients on a group of buildings.

A recent study, conducted by Tong *et al.* (2024), used a deep learning model to extend wind pressure coefficient time-series. The study showed that the deep learning model is superior to traditional methods (Correlation Distortion Static Transform and Gaussian Kernel Density Estimation) in constructing wind pressure sequence through comparing the statistical parameters, spectra, and distribution of extreme values.

Weng and German Paal (2022) and Meddage *et al.* (2024) employed decision tree-based regressors, specifically, the two studies utilized the Gradient Boosting technique for wind load regression. Gradient Boosting was used by Weng and German Paal (2022) to forecast wind pressure coefficients for non-isolated low-rise buildings using the TPU database. On the other hand, Meddage *et al.* (2024) employed Extreme gradient boosting (XGBoost) for the prediction of wind loads on tall buildings.

A limited number of studies have explored the application of ML to predict wind loads on non-rectangular buildings. This includes Circular cylinders (Hu and Kwok, 2020), C-shaped (or U-shaped) buildings (Mallick *et al.* 2020 and Meddage *et al.* 2021), and Y-shaped buildings (Sanyal and Dalui 2022). These studies primarily focused on predicting wind loads for specific single-building shapes, limiting the generalizability of the ML models to other architectural geometries. This study aims to address this limitation by incorporating four nonrectangular building

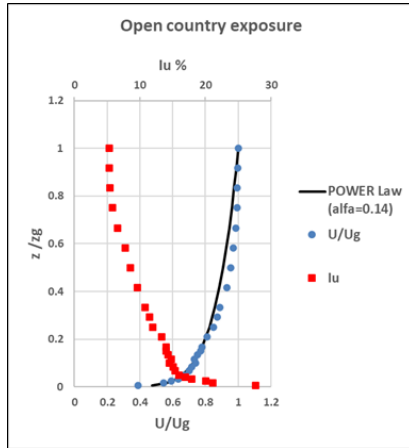


Fig. 5 Experimental wind speed and turbulence intensity profiles

shapes (L, U, T, and X) into a single ML model, developing a more generalized approach for predicting wind loads on non-rectangular buildings.

The data split approach influences the training process and the predictive performance of ML models. The random split often results in significant similarities between the training and testing sets, indicating implicit data leakage. The predictive performance of ML models trained and validated using sets selected randomly was found to be overestimated by Plonton *et al.* (2020) and Sogaard *et al.* (2021). While a structured split (e.g., split based on wind direction) results in more divergent training and testing sets. This often provides a realistic evaluation of the model's effectiveness in testing.

This paper investigates wind loading on roofs of nonrectangular buildings with L, U, T, and X shapes and highlights the differences between wind loads received by rectangular and nonrectangular roofs. The paper also implemented GBR and ANNs to predict wind pressure coefficients on nonrectangular buildings using the experimental dataset, generated at the wind tunnel. Two approaches for data split were used: random split and structured split. In the following sections, the paper starts with a description of the experimental methodology and wind tunnel testing. Then the wind results are presented in Section 3. Subsequently, the study presents the data preprocessing and provides a detailed explanation of the ML models, training methodology, and the predictive performance of the ML models. Finally, the key findings are presented in the last section.

2. Experimental methodology

This section describes the wind tunnel methodology, including the model configuration and the simulation of the atmospheric boundary layer. Based on the survey presented previously, the common nonrectangular shapes are L, U, T, and cross shape (X). Therefore, it was decided to investigate wind loads on nine nonrectangular models: three L-shaped models (L1, L2, and L3), three U-shaped models (U1, U2, and U3), two T-shaped models (T1 and T2), and

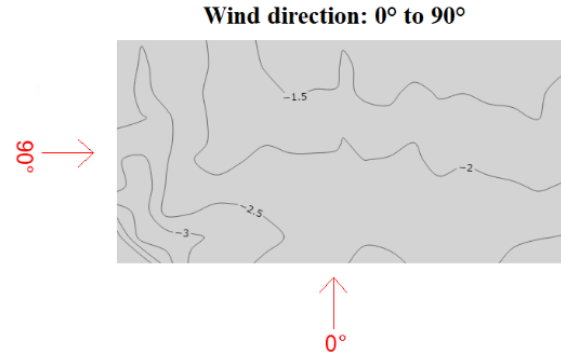


Fig. 6 Contours of enveloped negative peak pressure coefficients for a rectangular roof (30 m × 15 m)

one X-shaped model (X). Fig. 3 presents plans of the nine configurations tested in the wind tunnel (WT). It can be observed that the inner edges are marked in blue, and the outer edges are in black. This would serve as a reference for the discussion that will be presented in Section 3.

The construction of wind tunnel models is known as both costly and time-consuming. For irregular shapes, fabricating numerous models can be impractical. As a solution, irregular shapes were formed by combining a base rectangular model, equipped with pressure taps, with dummy models. The study employed a base rectangular wind tunnel model with a geometric scale of 1:200, see Fig. 4. The base model incorporated 193 pressure taps, distributed quasi-uniformly across the roof surface, to accurately trace wind pressures all over the roof zones. The dummy models were combined with the base model to produce the irregular configurations, (L, T, U, and X); the idea is to form the irregular models by putting side-by-side the base and dummy models several times until the entire configuration is covered. Fig. 4 shows an example of combining the dummy and the base models to form the L-shaped model. Testing of the L-shaped model did not require moving the base model several times, as two-thirds of the roof was covered by one test, and the wind pressures were concluded on the rest of the roof by symmetry. Similarly, scanning of wind pressures on the X-shaped model was performed through one test. On the other hand, two wind tunnel tests were conducted for each of L2, L3, and the T- and U-shaped models; the base model was moved twice to cover the entire roof.

Most wind tunnel findings were reported to describe wind loads on rectangular buildings. Therefore, it was sufficient to test the building model from 0° to 90° to determine the worst wind pressures on one-quarter of the building and conclude the wind pressures on the other three quarters based on symmetry. In the case of the nonrectangular buildings tested by combining base and dummy models, it was required to test the model for a full circle (0° to 360°) to reveal the changes in wind pressures all over the building envelope. Consequently, the L-shape models underwent wind direction testing with a range of (0°-355°) at a step of 5°, while the shapes of U, T, and X were tested from 0° to 350° with increments of 10°. Furthermore, the full-scale equivalent testing height was 10 m for all tests.

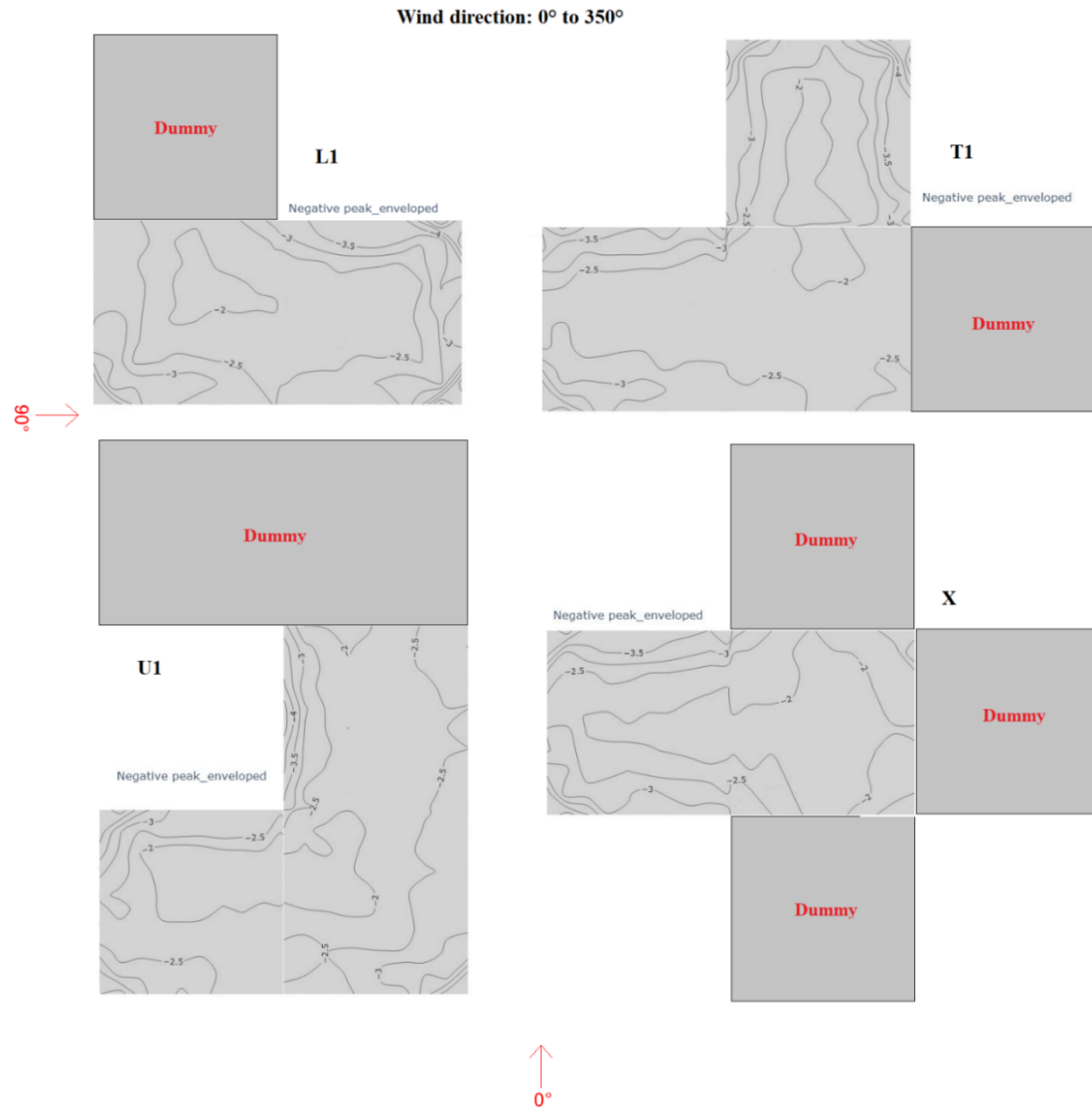


Fig. 7 Contours of enveloped negative peak pressure coefficients

The simulation of the atmospheric boundary layer was performed at the building aerodynamics laboratory of Concordia University. The tests were conducted in a simulated open country exposure as illustrated in Fig. 5, which presents the experimental wind speed and turbulence intensity at the testing section. The estimated power law exponent of the wind speed profile is 0.14. The estimation of mean and peak pressure coefficients in this study was done by normalizing the measured wind-induced pressures by the dynamic pressure at the roof height. The peaks were evaluated after extreme value analysis. Further details on wind tunnel characteristics, simulated wind profile, pressure measurements, and data processing can be found in (Stathopoulos 1984, Aldoum and Stathopoulos 2020, Chavez *et al.* 2022, Chavez *et al.* 2023, and Aldoum *et al.* 2023).

3. Wind tunnel results

This section starts with presenting contours of

enveloped negative peak pressure coefficients for a roof of a rectangular building, as shown in Fig. 6, and for a wind azimuth range of 0° to 90°. The rectangular building is 30 m × 15 m in plan and was tested in the wind for a height of 10 m and the same terrain conditions. Fig. 6 indicates that the highest local peak was developed at the corner with a value of around -5, while the edge zone experienced less section with peaks ranging from around -2.5 to -3.5.

Fig. 7 presents the local enveloped wind load distributions for four nonrectangular models (L1, T1, U1, and X). Unlike Fig. 6, these models were tested for a wider wind direction range (0° to 350°). Additionally, Fig. 7 includes areas where pressure can be inferred by symmetry, indicated as "dummy". The contours indicate that the maximum experimental peak experienced by the outer edges of the L configuration is around -3, which is comparable to peaks developed on the rectangular roof edge. However, the inner edges experience higher suction, exceeding -4. In addition, the distribution of peaks greater than -3.5 extends over almost the entire inner edge. On the other hand, the local peaks are with value of almost -5 on

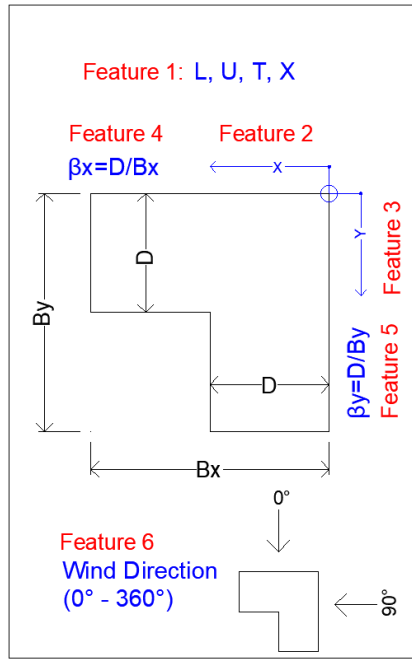


Fig. 8 Input features (variables)

the left outer corner and a bit greater than -5 on the right short-edge corner, therefore, it can generally be said that the outer corners of the L1 configuration receive loads similar in value to those experienced by the rectangular roof corner. The inner corner, however, experienced a higher peak of around -6, and relatively higher peaks cover the entire inner corner. This also applies to the short-leg inner corner of L2, in which the loads are higher than those of the outer corner and the corner of the rectangular building. For the other two L configurations, L2 and L3, similar conclusive observations were made when comparing the local wind loads on the inner and outer edges, and edges of the rectangular building. However, the inner corners of L3 and the inner corners on the long leg of L2 experience local peaks similar to those of the outer corners and the rectangular buildings.

Similar observations were also made for the T- and X-shaped configurations; the pressure values and distributions on the inner edges and corners are higher than loads on the outer edges and corners and those of the rectangular building. This is, of course, expected, at least for the local pressures, as the L-, T-, and X-shaped buildings share the same aerodynamic geometry on the inner edges side. Regarding the U-shaped configuration, building edges exhibit a pressure distribution trend that bears resemblance to the L shape; however, it experienced even higher suctions of up to -5 on the inner edges as shown for the U1 contours, this is likely attributed to the increased turbulence of the flow in the recess of the U-shaped building. For the corners of the narrow recess U-shaped buildings, U1 and U3, it was found that the local peaks of the inner and outer corners, and the corner of the rectangular building are generally the same with a magnitude of around -5, while for the wide recess U configuration, U2, the peaks of the inner corner are higher than the outer corners peaks and those of the corner of the rectangular building.

One-hot encoding				
Feature 1	Feature 1			
L	1	0	0	0
T	0	1	0	0
U	0	0	1	0
X	0	0	0	1

Fig. 9 One-hot encoding of the four categories

4. ML training and testing

This section provides details on the methodology utilized to transform and input the experimental data into the machine learning system and hyperparameter optimization of the Gradient Boosting regressor and the ANNs. Gradient Boosting (Friedman 2001) was used for regression of the wind tunnel data which uses several estimators (typically decision trees) in which each tree is constructed in a way to reduce the loss resulting from the previous one; in addition, the trees are fitted to predict the residuals instead of the target values.

Data preprocessing and ML model training, optimization, and validation were implemented in Python using the Scikit-Learn library. Scikit-Learn (Pedregosa *et al.* 2011) is a library in Python that provides a straightforward implementation of several ML algorithms with a simple interface for regression and classification. For the ANN, the study uses the TensorFlow Keras Sequential model (Abadi *et al.* 2015, Chollet 2015) to fit the experimental data.

4.1 ML input features and samples

As mentioned previously, four irregular shapes are included in this study, L, U, T, and X, in addition, each shape is characterized by a specific arrangement of pressure measurement points, and different plan dimensions, and tested for several wind directions. The experimental dataset needs to be organized and translated into informative features (or variables) that can be input into the ML algorithm for training.

Fig. 8 shows six variables that could be read from the tested wind tunnel models and will be used as input features during training and testing. Feature 1 represents the shapes of the irregular buildings included in this study (L, U, T, and X), this is a categorical feature and needs to be converted to numbers through an encoder as will be explained in the next subsection. Features 2 and 3 tell the pressure measurement point on the roof through a distance in x and y , respectively. In addition, the nine models, as shown in Fig. 3, have different plan dimensions, this can be described by the aspect ratio in x , feature 4 ($\beta_x = D/B_x$), and the aspect ratio in y , feature 5 ($\beta_y = D/B_y$). In addition, feature 6 represents the wind directions (WD). It should be mentioned that the target variables are mean pressure coefficients ($mean C_p$), and negative peak pressure coefficients ($peak C_p$). Therefore, two different ML systems have been considered for the prediction of each target. Furthermore, the experimental dataset includes a total of 195,000 samples.

Original data

Feature 1				Feature 2	Feature 3	Feature 4	Feature 5	Feature 6	Target (Cp)
L				0.6	0.6	0.5	0.5	30	-1.4203271

Transformed data

Feature 1				Feature 2	Feature 3	Feature 4	Feature 5	Feature 6	Target (Cp)
1	0	0	0	0.02	0.02	0.5	0.5	0.083	-1.4203271

one-hot encoding

Normalization

No change

Normalization

No change

Fig. 10 Example of features transformation (one sample)

4.2 Data preprocessing and evaluation metrics

Data preprocessing is the most important step in ML as it enhances the ML model's performance significantly. The six input features have different ranges and data types. Feature 1 consists of four text categories. To convert these categories into numerical values suitable for machine learning algorithms, one-hot encoding was employed. As recommended by (Géron 2019), this technique is particularly effective for categorical features with less than 10 categories. In one-hot encoding, each category is represented by a separate binary feature. The value is set to 1 when that category is present in the data sample, and 0 otherwise, as shown in Fig. 9.

The remaining numerical features (Features 2, 3 and 6) possess different scales, which can hinder machine learning model performance (Géron 2019). To address this, features were normalized to a common range between 0 and 1. Normalization utilizes the maximum value for each feature to achieve this scaling. This ensures all features contribute equally to the model's learning process. It's important to note that Features 4 and 5 already fall within the desired range (0-1) and therefore do not require scaling. Furthermore, the target values typically do not necessitate scaling (Géron 2019).

Fig. 10 presents an example of feature transformation for one sample; the fourth and fifth features and the target remain unchanged, the first feature was transformed using one-hot encoding, and features 2, 3, and 6 were normalized by the maximum values in each. It should be noted that the categorical feature was converted into four features after encoding, resulting in a total of nine input features.

A variety of metrics are employed to assess the performance of machine learning models. These include measures such as the coefficients of determination (R^2), mean square error (MSE), and other metrics to evaluate particular performance objectives of the model. This study utilized four metrics to evaluate the performance of the GBR and ANN: R^2 , MSE, mean absolute error (MAE), and mean bias error (MBE). The mathematical formulas of the four metrics are shown in Eqs. (1)-(4).

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - y_p)^2}{\sum_{i=1}^n (y_i - y_{mean})^2} \quad (1)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - y_p)^2 \quad (2)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y_p| \quad (3)$$

$$MBE = \frac{1}{n} \sum_{i=1}^n (y_p - y_i) \quad (4)$$

Where y_p is the predicted value, y_i is the actual value (evaluated based on the wind tunnel data), y_{mean} is the mean of y_i , and n is the number of samples within the selected set (e.g., training, validation, testing). R^2 reflect the proportion of variance in the target values (pressure coefficients) that is explained by the prediction of the ML model. MSE is used to quantify the mean squared difference between the target values and the ML model predictions. This metric squares the errors; thus, it signifies the large errors. MAE, on the other hand, calculates the average of the absolute values of errors; it is less sensitive to large errors compared to MSE. MBE is a reflection of the bias in the ANN predictions, a positive MBE indicates overestimation, and a negative MBE reflects underestimation of the wind tunnel pressure coefficients.

4.3 Gradient boosting regressor (random and WD-based splits)

To optimize the performance of the GBR, a technique called GridSearchCV with 4-fold cross-validation was employed (Pedregosa *et al.* 2011). GridSearchCV systematically explores a predefined grid of hyperparameter values (parameter space) and selects the combination that yields the best performance on the validation set during cross-validation. In this case, we used 4-fold cross-validation, where the data is split into four folds. The model is trained on three folds and evaluated on the remaining fold, repeating this process for all four folds. This approach helps mitigate overfitting and identify the hyperparameter combination that generalizes best to unseen data.

The parameter grid or space is a set of parameter ranges that includes a potential optimal value for each parameter. Constructing the parameter grid for hyperparameters optimization is challenging and could be time-consuming during runtime. Ideally, the parameter grid for a certain model should include all parameters associated with that model in all possible ranges. However, expanding the search space is computationally heavy and requires a long run time.

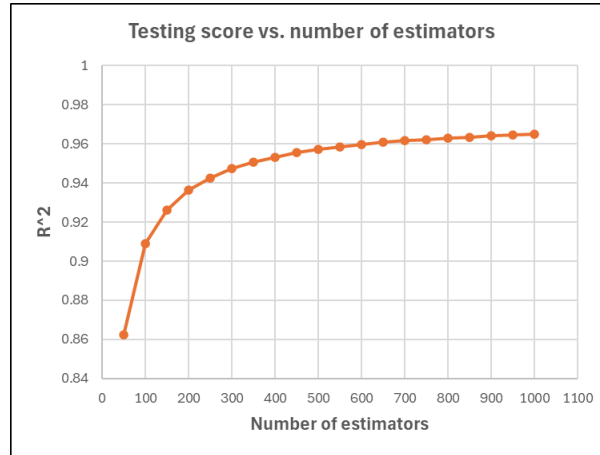


Fig. 11 R-squared values versus number of estimators

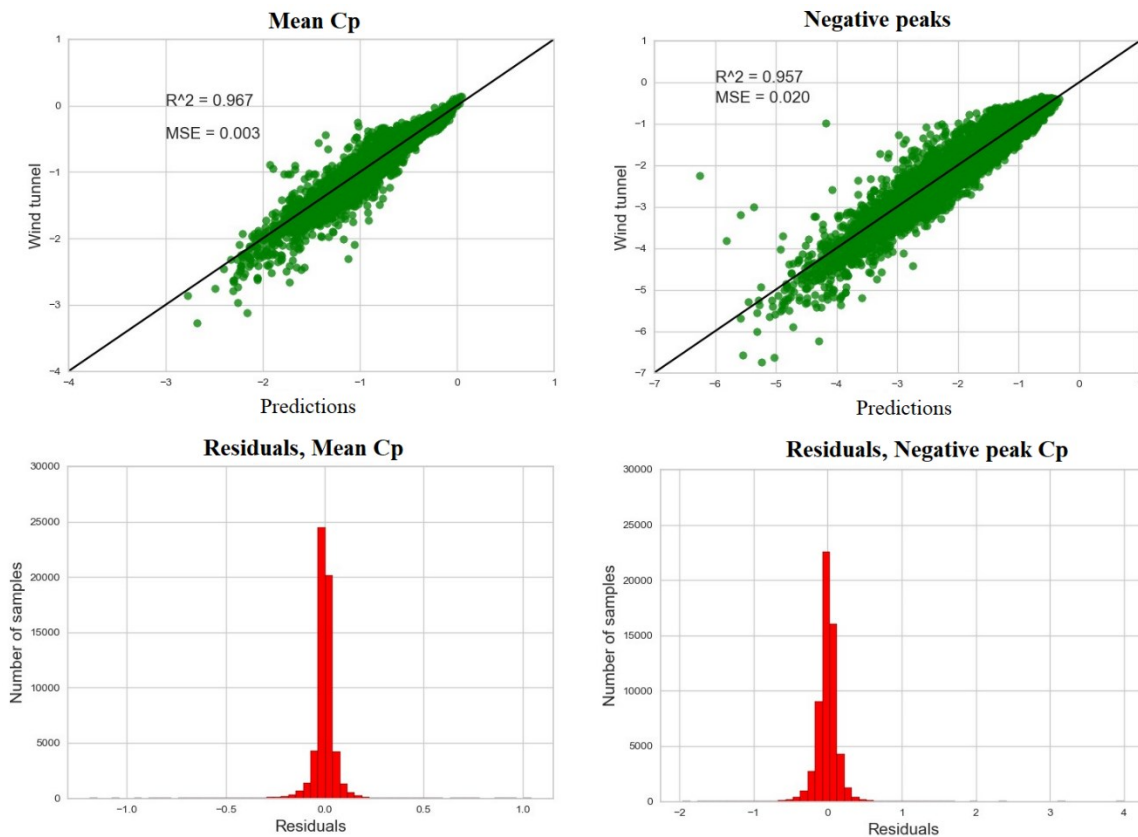


Fig. 12 Comparison of wind tunnel pressure coefficients and predictions of GBR on the testing set, random split

Given the relatively huge number of hyperparameters of GBR, including every hyperparameter and with a wide range, is impractical. Therefore, the best practice is to select a few hyperparameters with a limited range and implement the search, if the validation score on the testing set is acceptable, then the parameter grid is excellent, otherwise, the grid should be updated by increasing the range of the hyperparameters or adding extra parameters to the grid (in some occasions, both actions should be applied).

The grid search was performed on four hyperparameters: number of estimators, max_depth, max_features, and min_samples_split, and the initial training/testing split ratio was 7:3, the split was performed randomly. The training set

was then used for hyperparameters optimization and training of the GBR algorithm, and the unseen testing set was dedicated as a validation set of the ML models and evaluate their performance and efficiency. The results of the grid search indicated that the optimal values of max_depth, max_features, and min_samples_split were 10, 6, and 8, respectively. Regarding the number of estimators, the optimal value was 1000 based on the grid search. However, further investigation was performed by evaluating the testing score for GBR with different numbers of estimators in the range of 50 to 1000 at a step of 50, while the other hyperparameters were assigned to the optimal values. The variation of the GBR prediction accuracy for the testing set

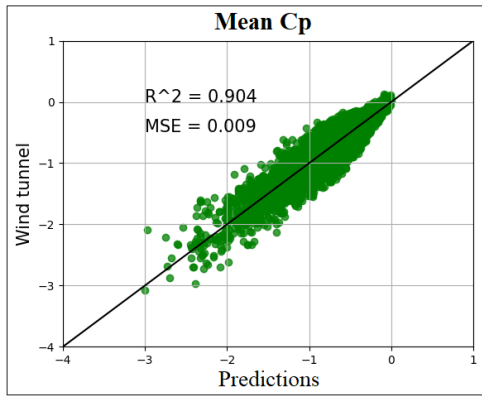


Fig. 13 Comparison of wind tunnel pressure coefficients and predictions of GBR on the testing set, WD-based split

was then plotted against the number of estimators, n , in Fig. 11. It is evident that the accuracy is proportional to the number of estimators used in the GBR model; however, the R-squared value is marginally improving after $n = 500$ at which the coefficient of determination is about 0.960. This indicates that increasing the number of estimators beyond 500 will only add extra complexity to the model without gaining a significant improvement in the prediction accuracy. Therefore, a value of 500 estimators was deemed optimal for GBR.

The size of the testing set greatly influences the confidence in the validation and robustness of a machine-learning model. A larger testing set enhances confidence in the model's predictions. However, a larger testing set *also* means less data available for training the model to understand the dataset's patterns and relationships. Therefore, finding the right balance between the size of the testing set for validation and the size of the training set for model comprehension is crucial for effective machine learning model development. Searching for the balance requires trying several testing set sizes and adopting the optimal size for the testing data. The validation R-squared and MSE values of the peak C_p for a range of testing data size; 0.1–0.9 at a step of 0.1, were calculated. It was found that the performance of the model on the testing set increases as the testing set size decreases. However, the performance was marginally improved when the size of the testing set was reduced from 0.3 to 0.2; the R-squared values increased by approximately 0.42%, changing from 0.957 to 0.961, and MSE changed from 0.02 to 0.018. Therefore, it was decided to use a test-to-train ratio of 3:7 as the resulting enhancement of the model outputs is small.

Fig. 12 presents the performance of the GBR for predicting both mean and negative peak pressure coefficients (C_p). The GBR predictions are compared to the wind tunnel pressure measurements. The figure also includes histograms displaying the distribution of the residuals (errors) between the predicted and measured values. The GBR achieved remarkable accuracy, evidenced by R-squared values of approximately 0.96 for both mean and negative peak C_p . In addition, the evaluated MSE of the mean C_p and negative peak C_p are 0.003 and 0.02, respectively, signifying negligible overall error.

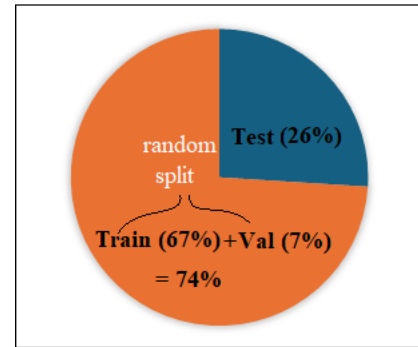


Fig. 14 Splitting of the dataset into training, validation, and testing sets

Furthermore, the histograms reveal an interesting trend. For the mean C_p , most instances in the testing set exhibit residuals close to zero, suggesting minimal discrepancies between predictions and actual measurements. However, the residuals for negative peak C_p tend to be larger, likely due to the inherent wider range of negative peak C_p values compared to the mean. Nonetheless, the histograms revealed a few instances where the error is relatively large, indicating poor performance by the GBR in predicting those particular cases.

Although Fig. 12 indicates an excellent overall performance on the testing set; however, the unseen set was selected randomly, resulting in high similarity between instances within the testing set and the training set, which suggests implicit data leakage. This can lead to an overestimation of the true performance of the GBR to generalizing to the experimental data. In addition, the random split does not allow for efficient performance inspection. For example, if a pressure coefficient contours map (L1 roof) is presented for a certain wind direction, then the reader would rely on 30% of the contours to judge the model's efficiency, as the remaining 70% were seen by the model during training.

To investigate these issues, the methodology was changed, and the dataset was split based on WD, a structured split. Twelve WDs (20° , 50° , 80° , 110° , 140° , 170° , 200° , 230° , 260° , 290° , 320° , and 340°) were set aside to serve as a testing set and the training set was formed by including the rest of the wind directions. This approach enables a comprehensive evaluation of the model's efficiency using full roof contour maps (and other graphs) generated exclusively from the unseen set. Although the similarity still exists, the two sets are more divergent than in the random split.

GBR was utilized again, using training and testing sets (split based on WD) and implementing the same grid search utilized previously. Its performance, however, on the new varied testing set yielded a decreased level of accuracy, with a maximum R-squared value of around 0.9 (for mean C_p), as shown in Fig. 13. This confirms the implicit data leakage discussed above, which results in overestimation of the evaluation score of the GBR. The decreased score of the GBR suggests that a better model is required to enhance predictive performance and provide more accurate results. Consequently, a Feedforward Neural Network (ANN) was

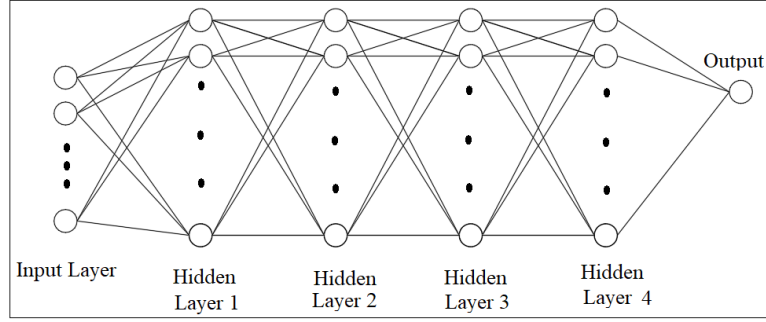


Fig. 15 Architecture of the ANN

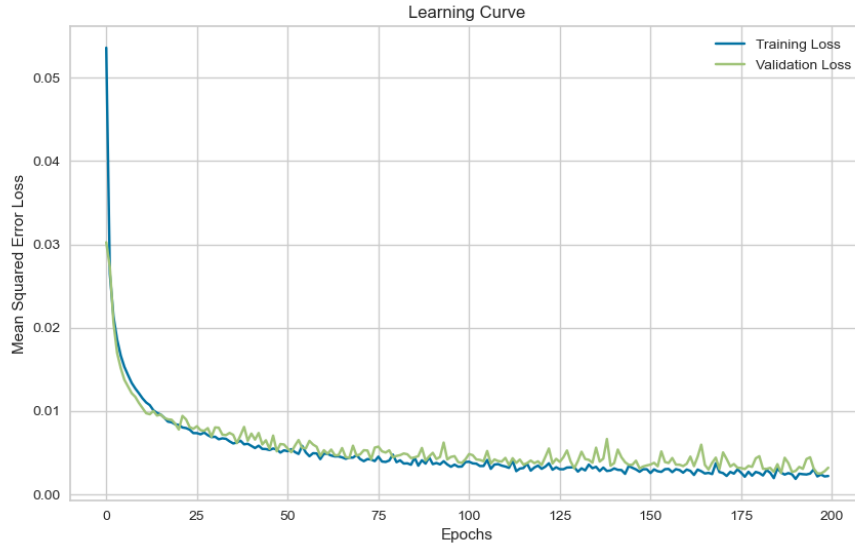


Fig. 16 The learning curve of the mean Cp ANN

then implemented as presented in the following subsection.

4.4 Artificial neural network

The TensorFlow Keras sequential model includes three types of layers: input, hidden, and output layers. The input and output layers consist of nine neurons and one neuron, respectively. This is so because the input and output layers are governed by the shape of the dataset, nine input features, and one target (mean Cp or peak Cp). The number of hidden layers and the number of neurons in each hidden layer were determined by experimentation; after evaluation of several configurations, it was found that four hidden layers with a number of neurons of 100, 500, 500, and 100 provide satisfactory results, and consequently, it was adopted. The dataset was split into three sets: training, testing, and validation sets. The training and validation sets are dedicated to the training phase. First, the data was split into testing and training sets based on wind directions as described previously; the testing set was chosen to include 12 wind directions while the rest of the wind directions were chosen to be included in the training set, this results in a split ratio of about 26% (testing) to 74% (training). 10% of the training set was randomly selected to create a validation set, which is equivalent to about 7% of the entire

dataset. Fig. 14 presents a chart that includes the percentage of each set with respect to the entire dataset.

Fig. 15 shows a schematic of the architecture of the ANN. In a fully connected sequential model, the neuron in a single layer receives an input, z , from the previous layer, the weighted sum of the outputs from all neurons in the previous layer plus a bias, which is represented in Eq. (5):

$$z = \sum_{i=1}^n (x_i w_i) + b \quad (5)$$

Where n represents the total number of neurons in the previous layer, i denotes the order of the neuron in the previous layer, x_i and w_i are the output and weight of the i th neuron, respectively, and b is the bias. The value z is then passed through an activation function to determine the final output value to the next layer.

The optimal parameters for the ANN were determined using an extensive search. The search identified the following as the best parameters: the Adam optimizer with a learning rate of 0.001, batch size of 200, and 200 epochs for training. Rectified Linear Units (ReLU) activation functions, which output the input directly if it is positive and zero otherwise, were set for the hidden layers, while a Linear activation function was applied for the output layer.

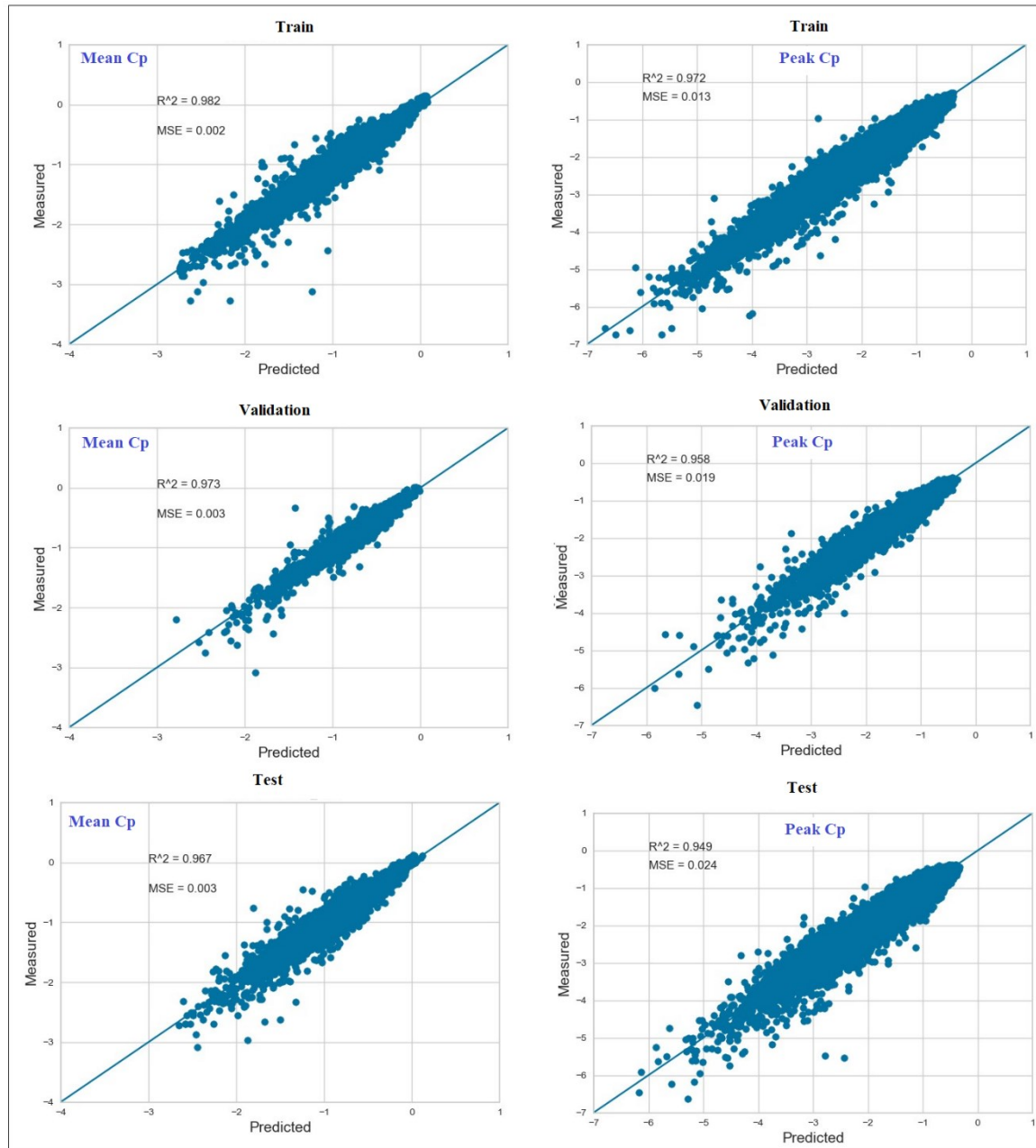


Fig. 17 Predictions of the ANN compared with the corresponding wind tunnel mean and peak C_p 's for the training, validation, and testing sets

The Linear activation function passes the input through without any transformation. The other hyperparameters of the sequential model were kept to their default values. The mentioned hyperparameter values were found to result in an excellent performance of the ANN model.

Fig. 16 presents the learning curve of the sequential model used to predict the mean C_p values. The figure indicates rapid improvement in the model within the first 25 epochs, the learning process then continues improving at a gradual and slower pace until the maximum number of epochs (200 epochs) is reached or is halted by the early stopping callback, which was set to prevent overfitting. For the peak C_p , a similar curve was achieved.

Fig. 17 compares the predictions of the ANN models

(predicted, horizontal axis) and the wind tunnel mean and peak pressure coefficients (measured, vertical axis) for the training, validation, and testing sets, the corresponding R^2 and MSE values are included in each plot. The ANN models of mean and peak C_p 's exhibited an excellent performance on the testing set (unseen data). This is evidenced by the strong correlations between the measured and the predicted values, with R^2 values of 0.967 and 0.949 for mean and peak C_p 's, respectively. While a few instances from the unseen sets were predicted with relatively large errors, the majority of predictions exhibited high accuracy. Further, the peak C_p predictions, despite excellent overall performance, demonstrated lower accuracy compared to the mean C_p 's. This is probably attributed to

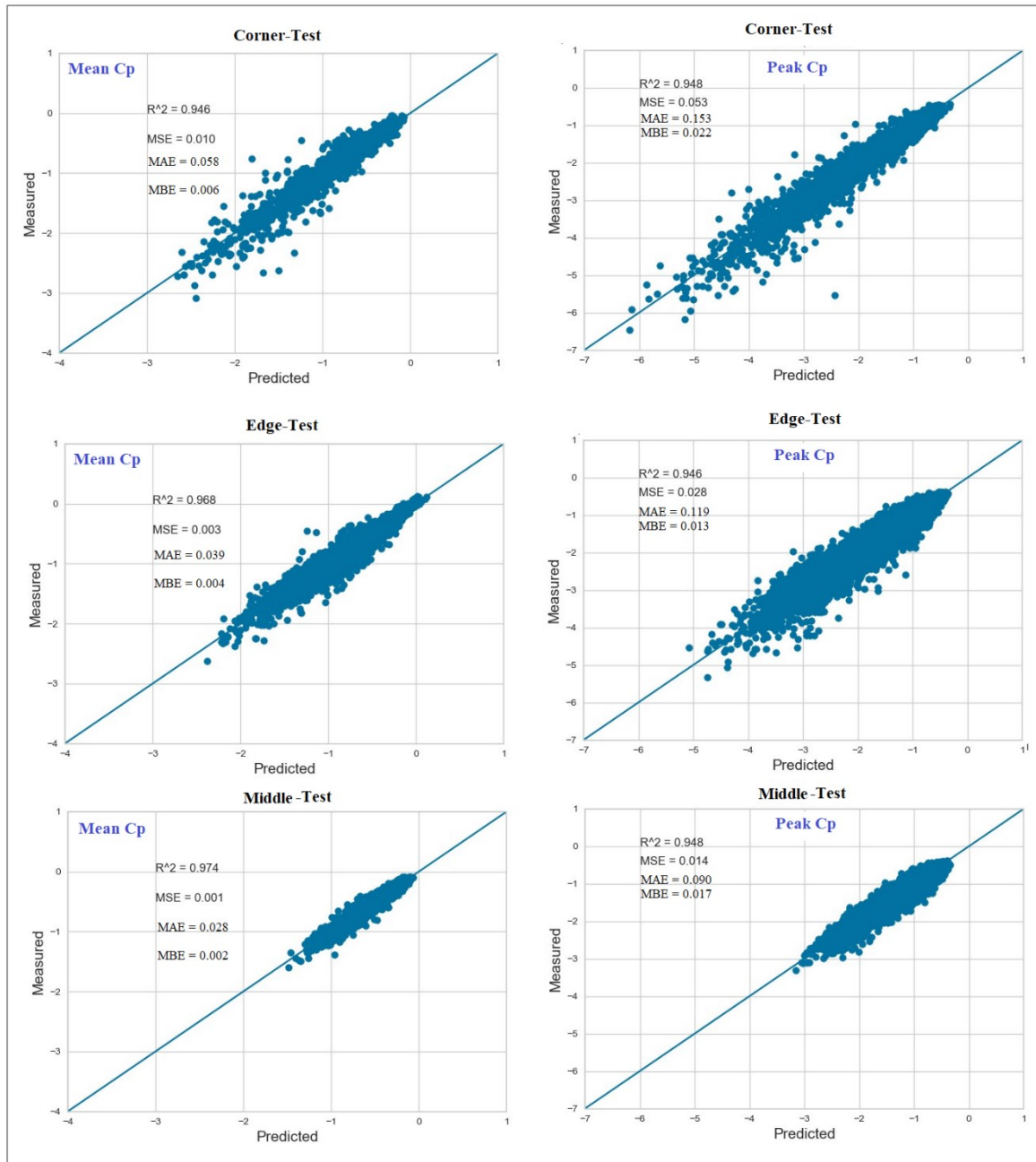


Fig. 18 Predictions of the ANN compared with the corresponding wind tunnel mean and peak C_p 's on the testing set for the three roof zones

the inherent complexity and variability of the fluctuating components of the wind pressures. The scatter plots and R^2 values evaluated for the training sets of both mean and peak C_p 's demonstrated an excellent ability of the ANN models to fit the training data, making strong connections between the input and the target variables without overfitting as indicated by the high performance on the testing sets. Similar accuracy was observed for the model's performance on the validation sets. The R^2 and MSE values evaluated for the three Mean C_p sets are comparable; however, those associated with the peak C_p sets exhibit a significant difference between training and testing errors, with testing MSE (0.024) nearly twice the training error (0.013). This indicates that fitting and predicting the mean pressure

coefficients was a simpler task compared to fitting and predicting peak pressure coefficients, confirming the complexity of the peak wind pressures.

Fig. 17 provided an overview of the accuracy of the ANN models in predicting mean and peak pressure coefficients across the entire testing dataset, without focusing on specific details such as the ability of the models to capture the critical roof pressures at the corners and edge areas of the nonrectangular roofs where the flow separation occurs, causing high suctions and complex pressure fluctuations. Therefore, it is important to investigate the degree of accuracy of the ANN models to map pressures in such areas. The testing set was further divided into three groups, each corresponding to taps located in the corner,

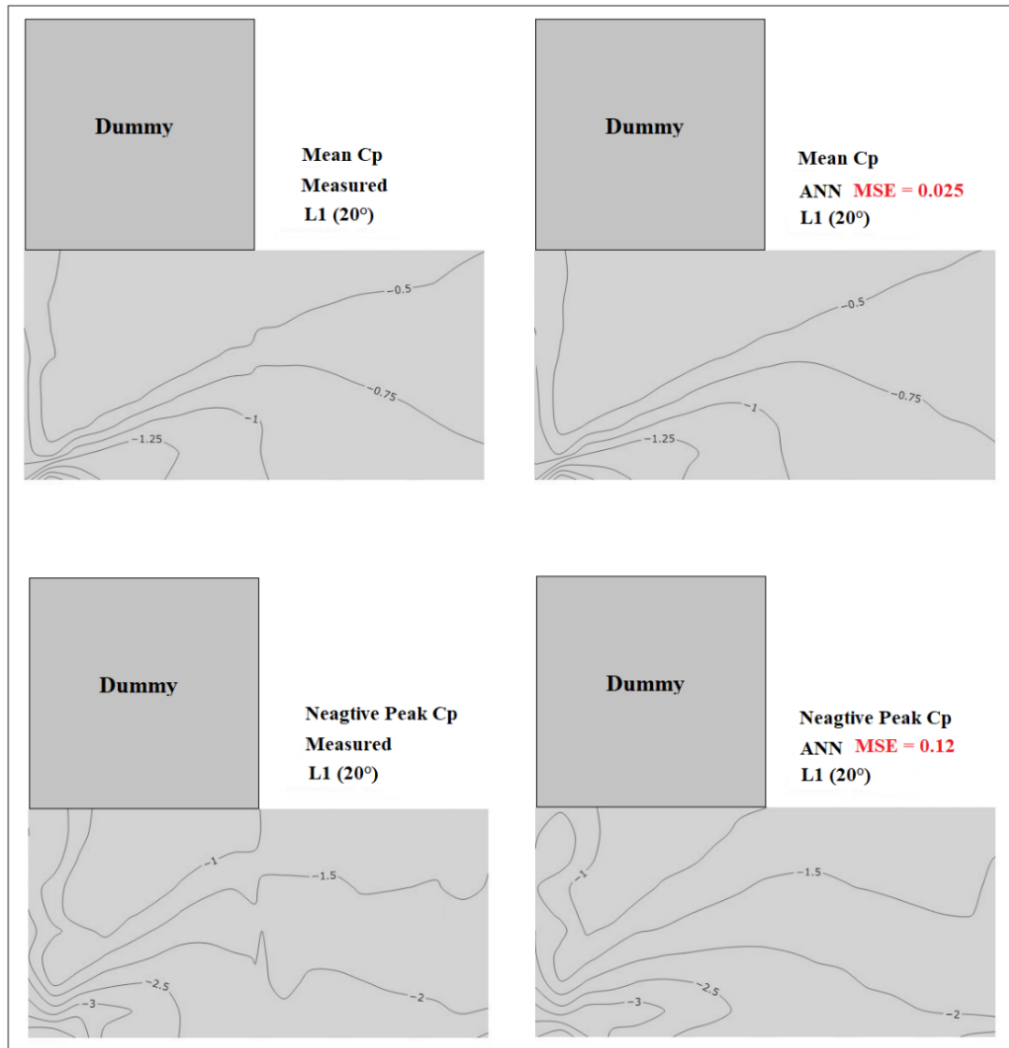


Fig. 19 Contours of mean and peak pressure coefficients, comparison between WT coefficients, and predictions of ANNs for the L1 model and 20° wind direction (testing set)

edge, and middle zones, as defined by NBCC 2020. Subsequently, the ANN models were used to predict the mean and peak pressure coefficients in these areas. The predicted values were then compared to the corresponding wind tunnel coefficients, as presented in Fig. 18. The figure indicates excellent performance on the ANN models in predicting the pressure coefficients on all zones. The lowest prediction accuracy was for the corner zone with an R^2 of around 0.95 and MSEs of 0.01 and 0.05 for the mean and peak C_p 's, respectively. This was expected as the corner zone is associated with the highest turbulent flow marked with the propagation of vortices due to the upstream flow detachment at the roof edges. In addition, the testing set was selected with cornering wind (i.e., oblique wind directions) or $\pm 10^\circ$ from the perpendicular wind directions, which makes the nature of the turbulent wind flow excessively chaotic at the corner. Nevertheless, the overall prediction accuracy of the corner zone pressures is satisfactory. The MAEs evaluated for the corner zone are 0.058 for the mean C_p and 0.153 for the peaks; these values are relatively higher than MSE's. This is so because most of the absolute differences between the WT C_p 's and predicted C_p 's are

less than one. The MBE, on the other hand, are 0.006 (mean C_p) and 0.022 (peak C_p), these values are relatively small, particularly the mean C_p biased error. In addition, the MBEs are positive, suggesting a slight overestimation of both mean and peak pressure coefficients by the ANN models.

The mean C_p ANN model demonstrated better performance in predicting coefficients within the edge zone compared to the corner zone, as evident in Fig. 18, with a coefficient of determination of around 0.97 and a significant reduction in both MSE and MAE for the edge zone predictions. For the peak C_p , the R^2 values are comparable for prediction within the corner and edge zones; however, the other evaluation metrics demonstrated improved accuracy for predictions within the edge zone. This case highlights the importance of considering multiple metrics for the assessment of model accuracy, as relying on one metric would, sometimes, result in a misleading conclusion. ANN models exhibited the highest predictive accuracy for mean and peak C_p values within the middle zone. This superior performance is likely attributed to the nature of the middle zone as a flow reattachment area, characterized by

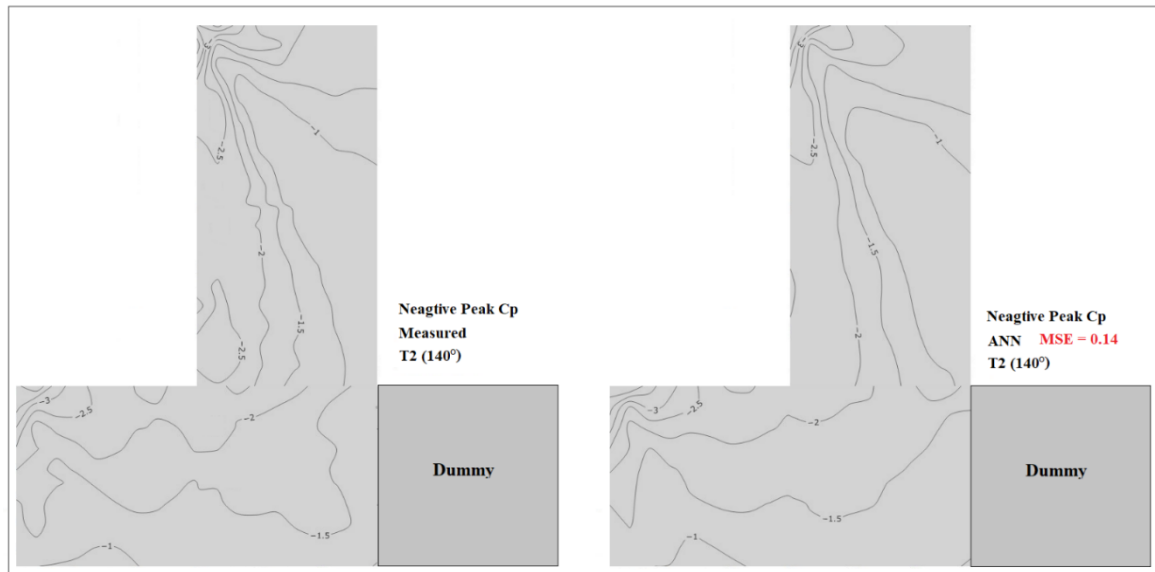


Fig. 20 Contours of peak pressure coefficients, comparison between WT coefficients, and predictions of the peak C_p ANN for the T1 model and 140° wind direction (testing set)

relatively smoother flow conditions and lower variability in wind pressure fluctuations. It should be noted that the MBE is positive for the middle and edge zones, indicating, as mentioned previously, a slight overestimation of the pressure coefficients.

To further investigate the ANN's performance on the testing set, Fig. 19 explores, in more detail, the ANN's performance on the testing set by visualizing the comparative distribution of mean and peak C_p values in contour plots for the L1 model at an angle of attack of 20° between the ANN predictions and the WT counterparts; the 20° angle of attack is one of the testing set wind directions. The dummy squares represent the areas that were not tested in the wind tunnel and can be concluded by symmetry. For the mean C_p 's, the ANN model mapped the entire roof area with minimal errors ($MSE = 0.025$), it also showed a slight overestimation of the bottom left area where the -1.5 -contour line relatively covers a larger area compared to the WT mean C_p distribution, confirming the positive MBEs discussed and presented in Fig. 18. The peak C_p 's exhibited similar trends; the overall accuracy is satisfactory and the overestimation, although small, is more pronounced compared to the mean C_p 's, particularly, for contour lines of -3 and -2.5 at the bottom left part of the plan. It is also interesting to note that the ANN model effectively captured peak C_p 's of more than -3 in the bottom right corner, despite its very limited presence in the WT peak distribution.

Fig. 20 shows a similar comparison but for the T-shaped model, T2, and 140° wind direction. The MSE is 0.14 , which is slightly higher than the corresponding average error evaluated for the L1 model peaks; this is probably attributed to the wind azimuth; the inner edges of the T shape are upstream, unlike the L1 scenario, where the inner edges are present downstream. This potentially made the wind flow over the T2 roof more complex, resulting in relatively more challenging trends of peak pressures to

generalize by the ANN model. The general distribution of peaks within the two contours is comparable, indicating that the peaks were reconstructed with high accuracy by the ANN. However, peaks at the left re-entrant corner, i.e., the inner corner at intersections between the T segments, show discrepancies; the wind tunnel contour lines indicate values with magnitude greater than -2.5 , and the ANN predicted suctions are lower than -2.5 .

Fig. 21 compares peak pressure coefficients predicted by the ANN and those evaluated based on the WT data for a line of taps on the roofs of L1 and U1, the figure includes the results of the 110° and 230° wind directions from the testing set, the peaks are plotted against a distance X normalized by the model width D . The peaks from the two sources exhibited an overall excellent similarity within L1 and U1 with few instances recorded errors of around 0.5 in the peak C_p values. These discrepancies appear on the windward side of the roof, as it is evident in the plot of the L1 peaks for X/D less than 0.1 when the wind azimuth is 110° and when X/D is between 0.8 and 1 for 230° wind direction. Similar results can be observed for the U1 model; however, the difference is smaller compared to the L1 model.

In summary, the ANN models exhibited excellent accuracy for predicting the mean and peak pressure coefficient, this was demonstrated by exploring the efficiency of the ANN models in predicting the mean and peak C_p 's across the entire testing set, the instances of the testing set within the roof pressures zones, the performance of the ANN models was also evaluated through roof pressure contours for wind directions within the testing set, and through a line of pressure taps along the roof. The predicted C_p 's were found to be highly correlated to the WT C_p 's with a coefficient of determination of up to 0.97 , and negligible errors were observed across most of the samples on the testing set.

Given the satisfactory predictive performance of the

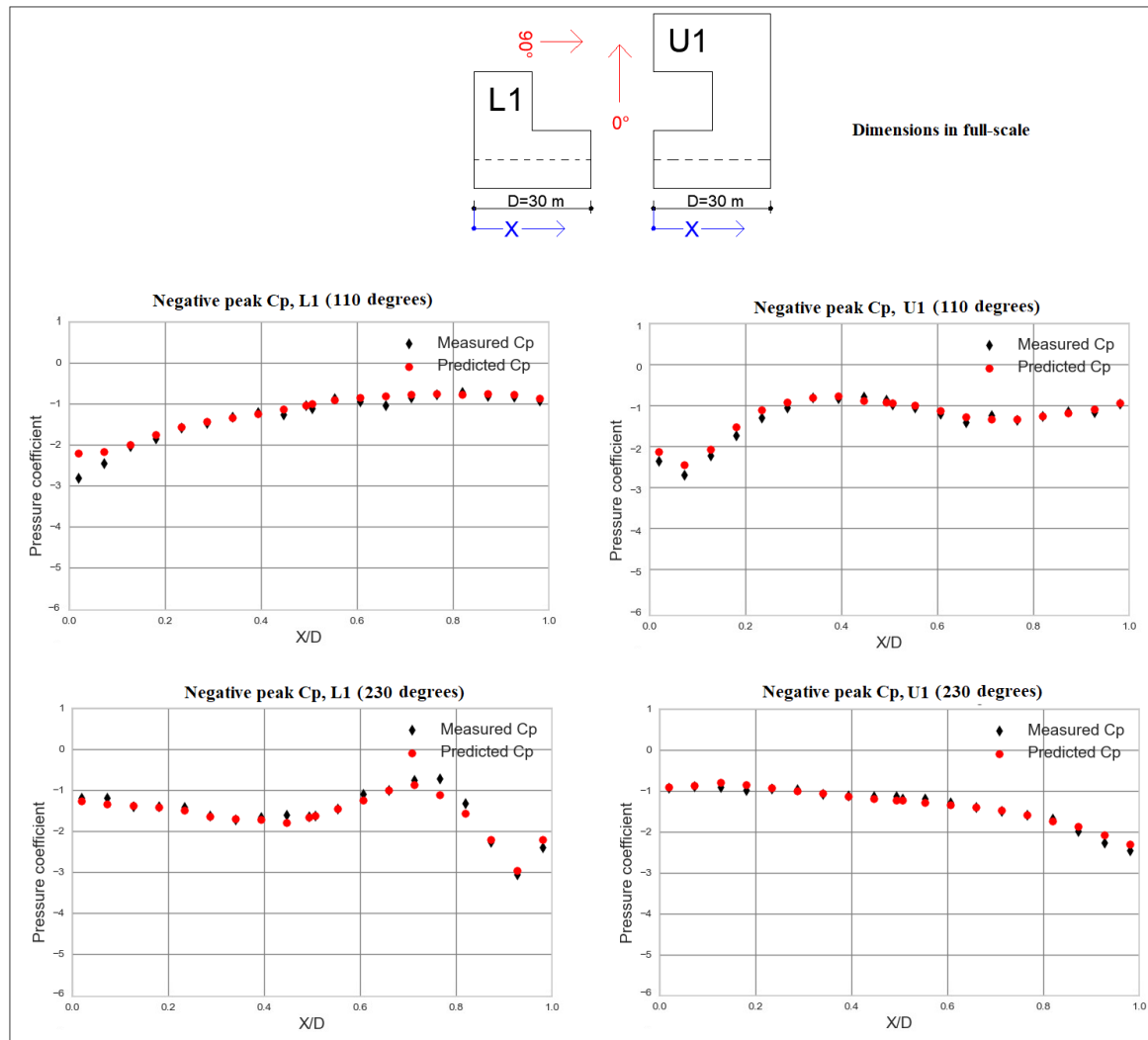


Fig. 21 Comparison between the ANN predictions of the peak C_p and the WT counterparts for a line of taps on roofs of L1 and U1

ANNs, the models can be utilized as a tool for estimating mean and peak pressures on roofs of nonrectangular buildings with the shapes of L, T, U, and X. This approach would significantly reduce the time, effort, and cost associated with testing nonrectangular models in boundary layer wind tunnels, providing a feasible tool for quantifying wind loads on roofs of such buildings.

5. Conclusions

In this study, wind loads were investigated on nonrectangular building roofs with shapes of L, U, T, and X through nine models. Additionally, wind tunnel data were utilized to develop Machine Learning models for predicting wind loads on such roofs, employing Gradient Boosting regressors and ANNs. The following are the study's key findings:

- The investigation into the local pressures indicated that the wind loads on the inner edges and corners are higher than the loads on the outer edges and corners and those experienced by rectangular buildings.

- GBR demonstrated excellent accuracy in predicting both mean and negative peak pressure coefficients, with evaluation scores of 0.96 based on the random split of the dataset. However, the GBR showed a reduced evaluation score (0.9) when the split was performed based on wind directions. The results from the two splits indicate that the random split often causes implicit data leakage, resulting in an overestimation of the performance of the ML models.

- The ANN models demonstrated an overall excellent accuracy in estimating wind pressures on nonrectangular buildings, with a coefficient of determination of up to 0.97 evaluated for the testing set and minimal errors. The models were found to be biased toward overestimating wind pressure coefficients; however, the bias was very small.

- Predicting mean C_p 's was slightly more efficient than predicting peak C_p 's, but the ANN model performed satisfactorily for both.

- The efficiency of the ANN models in predicting the pressure coefficients on the corner, edge, and middle zones is satisfactory, with the best performance achieved for the middle zone pressure coefficients.

- These ANN models provide practical and cost-

effective solutions for quantifying wind loads on roofs of non-rectangular buildings, saving a significant time and effort required for wind tunnel testing for non-rectangular buildings.

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